

CLASSIFICATION

Bayesian Classifier

- In some cases, the relationship between the attribute set and the class variable is non-deterministic
- **Bayesian Classifiers:** an approach for modelling probabilistic relationships between the attribute set and the class variable

Bayes theorem

Let X and Y be a pair of random variables

Joint probability, $P(X = x, Y = y)$

- Probability that variable X will take on the value x and variable Y will take on the value y

Conditional probability, $P(Y = y \mid X = x)$

- Probability that a random variable will take on a particular value given that the outcome for another random variable is known
- Probability that the variable Y will take on the value y , given that the variable X is observed to have the value x

Bayesian theorem

The joint and conditional probabilities for X and Y are related in the following way:

$$P(X,Y) = P(Y | X) \times P(X) = P(X | Y) \times P(Y)$$

Rearranging the last two expression, we get a formula, known as **the Bayes theorem**:

$$P(Y | X) = \frac{P(X | Y) \times P(Y)}{P(X)}$$

- 
- Consider football game between two rival teams: **Team 0** and **Team 1**
 - Suppose Team 0 wins 65% of the time and Team 1 wins the remaining matches
 - Among the games won by Team 0, only 30% of them come from playing on Team 1's football field
 - On the other hand, 75% of the victories for Team1 are obtained while playing at home
 - **If Team 1 is to host the next match between the two teams, which team will most likely emerge as the winner?**

- Let X be the random variable that represents the team hosting the match
- Y be the random variable that represents the winner of the match
- Both X and Y can take on values from the set $\{0,1\}$
- Probability Team 0 wins is $P(Y = 0) = 0.65$
- Probability Team 1 wins is $P(Y = 1) = 1 - P(Y = 0) = 0.35$
- Probability Team 1 hosted the match and Team 0 won is $P(X = 1 \mid Y = 0) = 0.3$
- Probability Team 1 hosted the match it won is $P(X = 1 \mid Y = 1) = 0.75$

Our objective is to compute $P(Y = 1 \mid X = 1)$, which is the conditional probability that Team 1 wins the next

$$\begin{aligned}
P(Y = 1|X = 1) &= \frac{P(X = 1|Y = 1) \times P(Y = 1)}{P(X = 1)} \\
&= \frac{P(X = 1|Y = 1) \times P(Y = 1)}{P(X = 1, Y = 1) + P(X = 1, Y = 0)} \\
&= \frac{P(X = 1|Y = 1) \times P(Y = 1)}{P(X = 1|Y = 1)P(Y = 1) + P(X = 1|Y = 0)P(Y = 0)} \\
&= \frac{0.75 \times 0.35}{0.75 \times 0.35 + 0.3 \times 0.65} \\
&= 0.5738,
\end{aligned}$$

$$P(X) = \sum_{i=1}^k P(X, Y_i) = \sum_{i=1}^k P(X|Y_i)P(Y_i).$$

- $P(Y = 0 | X = 1) = 1 - P(Y = 1 | X = 1) = 0.4262$
- Since $P(Y = 1 | X = 1) > P(Y = 0 | X = 1)$, Team 1 has a better chance than Team 0 of winning the next match

Using the Bayes Theorem for Classification

- Formalizing the classification problem from a statistical perspective:
- Let X denote the attribute set and Y denote the class variable
- If the class variable has a non-deterministic relationship with the attributes, then we can treat X and Y as random variables and capture their relationship probabilistically using $P(Y | X)$
- This conditional probability is also known as the Posterior probability for Y

- During the training phase, we need to learn the posterior probabilities $P(Y | X)$ for every combination of X and Y based on information gathered from the training data
- By knowing these probabilities, a test record X' can be classified by finding the class Y' that maximizes the posterior probability, **$P(Y' | X')$**

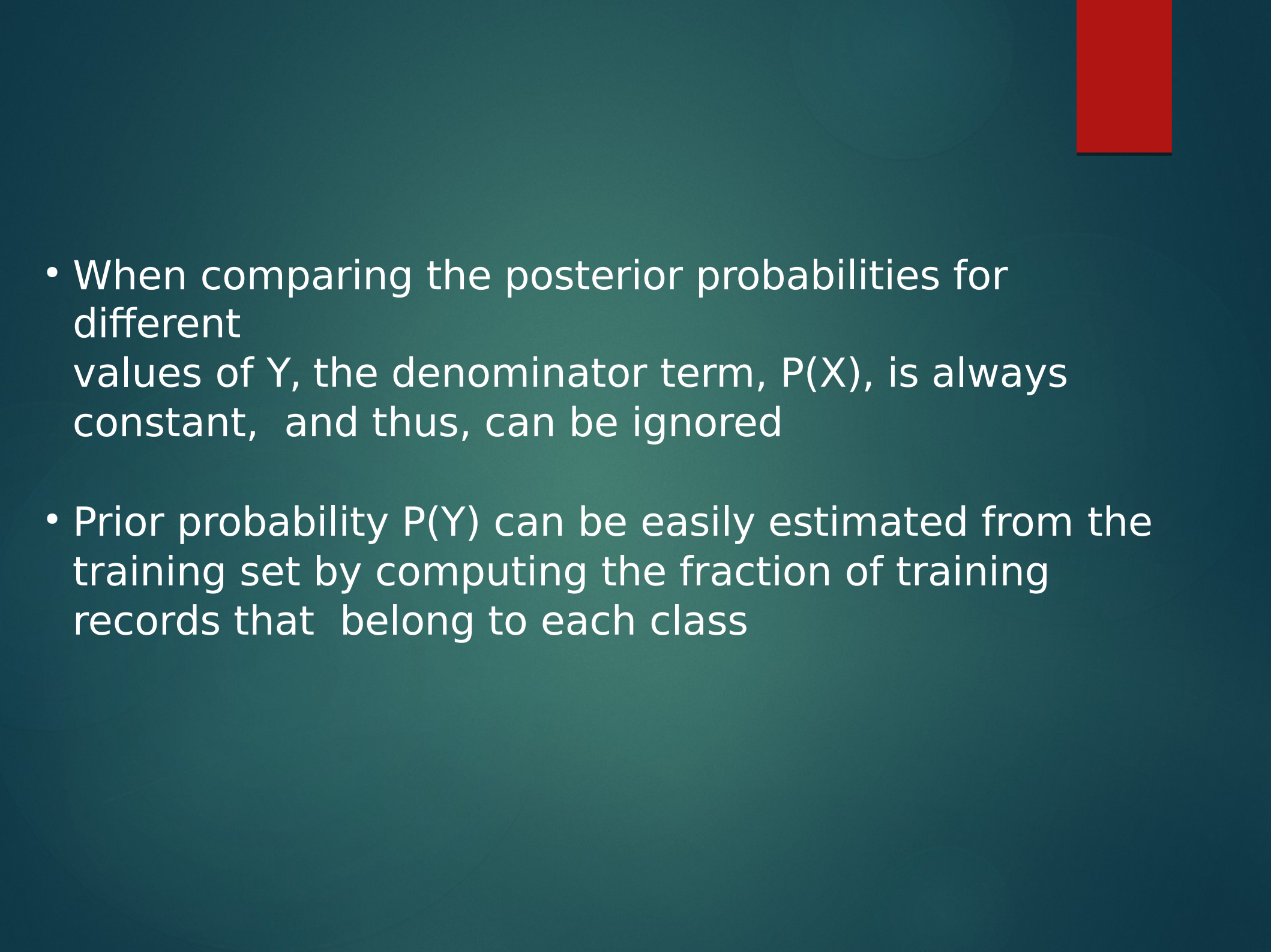
	binary	categorical	continuous	class
Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- **Example:**

- A test record is given with the following attribute set: $X = (\text{Home Owner} = \text{No}, \text{Marital Status} = \text{Married}, \text{Annual Income} = \$120\text{K})$
- To classify the record, we need to compute the posterior probabilities $P(\text{Yes} \mid X)$ and $P(\text{No} \mid X)$ based on information available in the training data
- If $P(\text{Yes} \mid X) > P(\text{No} \mid X)$, then the record is classified as Yes; otherwise, it is classified as No

- 
- Estimating the posterior probabilities accurately for every possible combination of class label and attribute value is a difficult problem because it requires a very large training set, even for a moderate number of attributes
 - The Bayes theorem is useful because it allows us to express the posterior probability in terms of the prior probability $P(Y)$, the class-conditional probability $P(X|Y)$, and the evidence, $P(X)$:

$$P(Y | X) = \frac{P(X | Y) \times P(Y)}{P(X)}$$

- 
- When comparing the posterior probabilities for different values of Y , the denominator term, $P(X)$, is always constant, and thus, can be ignored
 - Prior probability $P(Y)$ can be easily estimated from the training set by computing the fraction of training records that belong to each class

Evaluating the performance of classifiers

Holdout Method:

- The original data with labeled examples is partitioned into two disjoint sets- training and test sets
- Classification model is induced from training set
- Its performance is evaluated on the test set
- The accuracy of the classifier can be estimated based on the accuracy of the induced model on the test set

Limitations:

- Fewer labeled examples are available for training
- Model may be highly dependent on the composition of training and test sets
- Training and test set are not independent of each other

Evaluating the performance of classifiers

Random Subsampling:

- Holdout method can be repeated several times to improve the estimation of classifier's performance
- This approach is known as random subsampling

Limitations:

- Suffers from the problems associated with holdout method
- It also has no control over the number of times each record is used for testing and training

Evaluating the performance of classifiers

Cross Validation:

- Each record is used the same number of times for training and exactly once for testing
- K fold cross validation segments the data into k equal sized partitions
- During each run, one of the partition is chosen for testing, rest of them are used for training
- Procedure is repeated k times so that each partition is used for testing exactly once
- Total error is found by summing up the errors for all k runs

- Leave one out approach: special case where $k=N$, the size of the data
- Each test set contains only one record
- Advantages: Utilizes as much as data as possible for training
- Test sets are mutually exclusive and effectively cover the entire data set
- Drawback: computationally expensive to repeat the procedure N times
- Since each test set contains only one record, the variance of the estimated performance metric tends to be high